

ISSG Métier and transversal variables 2024

Background

The aim of the group is to maintain the list of DCF métiers on level 6 and level 7 that is used for data calls in ICES and STECF and to serve as a forum to discuss harmonization of methods used for assigning métier codes and working with transversal data (i.e. fishing activity data) in EU Member States.

The group has been ongoing since 2018, starting with a workshop discussing the methods used to assign métier codes to transversal data, issues and best practices, and the following years as an RCG ISSG, reports can be found [here](#). Achievements from the ISSG over the years have been:

- Implementation of [new standardized and harmonized list of métier codes on DCF level6](#) regionalized by RCG (and DCF level7 for LP fisheries since 2023), which was approved by RCG's in 2020 and in the September 2020 Liaison meeting, it was agreed by the NCs that the new codes for métiers and reference lists is used and implemented by the MS. [A table links between new and old codes](#) (in cases that a mesh size range has been split up, a choice has been taken to link to one of them).
- Reference lists:
 - [Reference species list on how to group species \(into group of species on DCF level5\)](#)
 - [Reference areas list establishing the link to RCG area](#)
 - [Reference gears list on how to group gears \(into gear DCF level4 and gears' group\)](#)
- [Script](#) that can assign métier codes using a specified data input format. It also has functionalities 1) to propose an estimate of métiers where all needed information is not available and 2) to refine the "rare" métiers firstly assigned by the general algorithm focusing on the year*vessel main métiers, in order to limit the multiplication of métiers calculated.
- [Manual](#) explaining the background, script, input format and reference lists
- GitHub repository ([RCGs/Metiers at master · ices-eg/RCGs \(github.com\)](#)) where all the material is available (reports, métier list, reference lists, script, manual)

The new métier codes are now being requested in ICES (*RDBES and VMS/Logbook*) and STECF FDI data calls.

In 2021, the group changed name to 'ISSG on Métier and transversal variable issues', also including a task to look at effort calculations for the small-scale fisheries (including the use of the fecR package) and a task to discuss harmonization of methods used to estimate fishing activity data in global (in particular to complete RDBES tables CE&CL). Since 2023, the group ensure continuity also with an initiative from RCG Econ to develop an alternative fleet segmentation in order to keep the link between the two approaches.

The ISSG is chaired by Sébastien Demaneche, Ifremer, France and Josefine Egekvist, DTU Aqua, Denmark.

Work-plan

ToRs and work plan (specific tasks) for 2023/2024.

1. Continue following and evaluating the implementation of the métier codes on a pan-regional level and maintaining métier codes and other reference lists and scripts.
2. Update métier descriptions from the 2023 RDBES data call (tables CE&CL). Consider the use of the métier description reports.
3. Based on the information received from the questionnaires sent out in spring 2023, evaluate the use of cross-validation methods in MS to combine data coming from different declarative sources, the ongoing standard practices and develop, on this basis, best practices guidelines, with a specific focus on the RDBES CE and CL tables.
4. Work on a template to document CE and CL data uploaded to RDBES
5. Continue following up on the development of the fecR package and its efforts to calculate fishing effort metrics that are harmonized/homogenized between MS
 - Collaborate with ISSG SSF regarding the inclusion of SSF in fecR
6. Continue following the development under RCG-Econ of an alternative fleet segmentation and advice on it in order to keep the link between the two approaches (depending on RCG-Econ work on this, especially the feedback of the 3rd workshop held in May 2023)

Progress during 2023/2024

The ISSG had the following online 1.5 - 2 hours meetings during the last year:

09-02-2024	Meeting to plan ISSG tasks for the 2023/2024 term.
04-03-2024	Meeting to follow up on ISSG tasks. Requests to add new codes were discussed and following invited experts participated in the meeting: Torsten Schulze (Germany), Mathieu Depetris (RCG LP), Hugo Maxwell (RCG LP), Charis Charilaou (RCG MBS)
05-04-2024	Meeting to follow up on ISSG tasks
22-04-2024	Meeting to finalize ISSG tasks and report

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Roadmap/follow-up

Main outcomes and communication between the ISSG and other groups

- Reviewed requests for new métier codes to add in the reference lists from countries and RCGs
- Created métier description reports from 2024 RDBES data call for 2021-2023 CL and CE data
- Analysed questionnaire on cross-validation methods on data sources, methods to combine different types of data and methods used to estimate value of landings to identify methods used and suggest best practices
- Suggestion for a template to document methods used for creating the RDBES CE and CL data.
- Tested fecR package with new version of R
- Made an overview of what is required in the EU-MAP table 6 and what is optional/mandatory in the existing Control Regulation Implementation Regulation.

Communication:

- For task 1, the ISSG chairs were in contact with ICES and JRC to update the métier codes reference lists (DCF level6 and level7) for data calls (e.g. ICES RDBES and STECF FDI data calls).

- For task 1, the ISSG chairs were in contact with the RCG LDF chair to answer a request regarding the added métiers on DCF level7 asked for LP fisheries
- For task 1, the ISSG chairs invited MS representatives to meetings if they requested on updating the métiers list.
- For task 4 communications with JRC on the GitLab that stores the fecR package code.
- For task 6, the ISSG chairs were in contact with the chairs of the RCG Econ fleet segmentation workshops to get an update on the status of the work.

Suggestions for the next step in intersessional work (future tasks)

1. Continue following and evaluating the implementation of the métier codes on a pan-regional level and maintaining métier codes and other reference lists and scripts.
2. Update and further develop métier description report, so that it can produce a multiannual report and can take the RDBES data license into account for making a public report.
3. Continue the analysis of questionnaires sent out in spring 2023, to evaluate the use of cross-validation methods in MS to combine data coming from different declarative sources, the ongoing standard practices and develop, on this basis, best practices guidelines, with a specific focus on the RDBES CE and CL tables.
4. Evaluate eventual feedbacks on the draft template elaborated in 2024 to document methodologies applied to inform RDBES CE & CL tables
5. Continue following up on the development of the fecR package to calculate fishing effort metrics that are harmonized/homogenized between MS.

Task 1: Continue following and evaluating the implementation of the métier codes on a pan-regional level and maintaining métier codes and other reference lists and scripts.

In 2023 new métier codes were requested in the STECF FDI data call and in ICES data calls (VMS/Logbook and RDBES). Within this task, requests for new métier codes to add and revision of other reference lists (area, gear or species) are evaluated, approved and implemented. The ISSG will also follow the implementation of these new developed métier reference lists in data calls. The task in 2024 focused on reviewing requests for new métier codes, discussing them during the ISSG meetings, and maintaining the métier reference lists according to decisions. The current métier reference lists (DCF level6 & level7) and other reference lists (e.g. species reference list) elaborated by the group are available in the following GitHub: <https://github.com/ices-eg/RCGs/tree/master/Metiers>

Request to update species reference list with “*Ammodytes marinus*”

A request to update the species reference list with “*Ammodytes marinus*” (QLH/Lesser sand-eel) from ASFIS 2022 was followed up, and a global review and update of the species reference list ('Metier Subgroup Species 2020 cleaned version.csv', file available in the GitHub) was made:

- From the RDBES CL data from 2021-2022 all species that have been reported (Landings, BMS and Discards registered) were listed.
- Based on this review, species that were missing in the species reference list were identified.
- For the missing species the same type of information as in the species reference list were defined (e.g Genus, EnglishName, TaxoCode, ISSCAAPcode, ISSCAAPdescription, Grouping_2) and added to the list associated to the combination 'FAOcode+ScientificName' related to the missing specie considered.
- Then the column "Grouping_3_DWS" was checked against the Deep water regulation (EC) 2016/2336 Annex I.
- The species '*Ammodytes marinus*' (QLH/Lesser sand-eel) that was requested to be added, was not reported in the previous RDBES CL data (2021 and 2022), but was added directly to the file.
- The output files were saved as '.rdata' and in '.csv', where the column names were slightly simplified for an easier R coding.
- Finally, the new species were added in the end of the original excel file and new version was uploaded on the GitHub. Then the group has a look into to validate the updates done. Furthermore, the species reference list on GitHub was renamed from 'Metier Subgroup Species 2020.xlsx' to 'Metier Subgroup Species.xlsx' and the reference to the species list in the métier R-script was updated.

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Request to update métier reference lists to ICES and JRC

Métier codes listed below have been agreed by the ISSG and have been added to the métier reference lists (level 6 and level 7) available on the GitHub. ICES and JRC have been then contacted to include these new métier codes in their métier reference lists for data calls (e.g. *ICES vocabulary for Metier6_FishingActivity* and *NationalFishingActivity for ICES RDBES data call*)

RCG area	Metier level 6 code	Description	Update
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MBS	LA_LPF_<14_0_0	Lamprette, Large pelagic fish, see the code reg. mesh size and selectivity device	Removed extra _0 from the code. The code replaces LA_LPF_<14_0_0_0
MBS	LA_SLP_<14_0_0	Lamprette, Small and large pelagic species, see the code reg. mesh size and selectivity device	Removed extra _0 from the code. The code replaces LA_SLP_<14_0_0_0
MBS	LA_SPF_<14_0_0	Lamprette, Small pelagic fish, see the code reg. mesh size and selectivity device	Removed extra _0 from the code. The code replaces LA_SPF_<14_0_0_0
MBS	LLS_LPF_0_0_0	Set longlines, Large pelagic fish, see the code reg. mesh size and selectivity device	Added on request from Malta
NAI	GNS_MPD_>0_0_0	Set gillnets, Mixed pelagic and demersal, see the code reg. mesh size and selectivity device	Added on request from Azores
NAI	GNS_MPD_100-119_0_0	Set gillnets, Mixed pelagic and demersal, see the code reg. mesh size and selectivity device	Added on request from Azores
NAI	GNS_MPD_120-219_0_0	Set gillnets, Mixed pelagic and demersal, see the code reg. mesh size and selectivity device	Added on request from Azores
NAI	GNS_MPD_>=220_0_0	Set gillnets, Mixed pelagic and demersal, see the code reg. mesh size and selectivity device	Added on request from Azores
LDF	OTM_CRU_>0_0_0	Midwater otter trawl, crustaceans, see the code reg. mesh size and selectivity device	Added on request from Norway
LDF	OTMCRU_16-31_0_0	Midwater otter trawl, crustaceans, see the code reg. mesh size and selectivity device	Added on request from Norway

RCG area	Metier level 7 code	Description	Update
MBS	LLS_LPF_0_0_0_SWO	Swordfish (<i>Xiphias gladius</i>)	Added on request from Malta

Below the ISSG discussions on requested métier codes are detailed also for the métier codes of which have not all been yet approved by the ISSG.

Request from Malta (Jurgen Mifsud). Charis (chair of RCG MBS joined the 2nd ISSG meeting as invited expert to discuss the request):

The group discussed the introduction of two new métier codes in the reference list requested from Malta.

- LLS_LPF_>_0_0_0
 - The group noted that the gear (LLS, level4) and target group of species (LPF, level5) codes are available in the EU-MAP commission delegated decision (EU 2021/1167), so they are in line with one of the ISSG establishing principles¹
 - Following ISSG establishing principles and for harmonization, the new métier code should be: LLS_LPF_0_0_0 (as LLS gear doesn't have a mesh size)².
 - **Following discussions, the group concluded and agreed to add, for the MBS RCG area³, the métier code LLS_LPF_0_0_0 to the métier DCF level6 métier reference list and the métier code LLS_LPF_0_0_0_SWO to the DCF level7 métier reference list (for LP fisheries needs).**
- GTN_MPD_>0_0_0
 - This requested métier code concatenates combined gears (*combined gillnets-trammel nets*) and mixed target group of species (*mixed pelagic and demersal species*)
 - At this stage, there is no métier codes in the reference list which concatenate combined gears and mixed target group of species as 1) the métier script will not be able to calculate it directly and 2) more details are preferred.
 - Nevertheless, combined gears and mixed target group of species have been included separately in some cases for some specific gears/fisheries in the métier reference list for national (regional) needs and could be specified directly in the input data for the métier script by countries which need it.
 - The question is also what is the intention of the fishery: is it a real mixed fishery, or do they target e.g., demersal fish and accidentally catch pelagic? The group argued that normally the principal target group of species in value or weight should be preferred exception of when real mixed group of species are targeted. A question remains also if it is small or large pelagic which are eventually combined with demersal fish?
 - The group noted also that the following métiers already exists in the métier DCF level6 reference list for RCG MBS: GNS_DEF, GNS_LPF, GNS_SLP, GNS_SPF & GTN_DEF with following mesh-size ranges "<16_0_0", ">=16_0_0" & ">0_0_0".

¹ Principles used for defining métier codes: 1. Gear-target species assemblage combinations (métier level 5) follow table 5 from EU-MAP commission delegated decision (EU 2021/1167), ...

² Principles used for defining métier codes: ... 4. "_0_0_0" for gears with no mesh size (e.g., longlines, hand lines, trolling lines), ">0_0_0" for unknown mesh size also for the following gears: traps, pots, beach seines and dredges (*gears with mesh size but for which no mesh size ranges have been defined*).

³ See metier codes "LLD_LPF_0_0_0" at DCF level6 and "LLD_LPF_0_0_0_SWO" at DCF level7 which already exist for MBS RCG

- The group noted that the gear (GTN, level4) and target group of species (MPD, level5) codes are available in the EU-MAP commission delegated decision (EU 2021/1167), so they are in line with one of the ISSG establishing principles⁴
- The group argued that, to seek consistency, if added, GTN_MPD métier code should be associated with the following mesh size ranges at DCF level6 “<16_0_0”, “>=16_0_0” & “>0_0_0” when the mesh size is unknown.
- **The group concluded that more information about this fishery is needed from Malta. This could be either provided for the ISSG to discuss, or to be discussed at the RCG MBS TM 2024. RCG MBS chair will contact Malta regarding this request.**

Request from Germany for two new gears/métiers codes. Torsten Schulze (from Germany joined the 2nd ISSG meeting as invited expert to discuss the request):

The group discussed the introduction of these two new gears codes in the reference lists.

- TBC used for Crangon/Brown shrimp fishery. Has a different impact on the sea floor than the TBB⁵. Germany would like to add the métier TBC_CRU_16-31.
 - The group highlights that this is not in line with the establishing ISSG principles⁶. The code TBC is not available in the EU-MAP commission delegated decision (EU 2021/1167) also not included in the ICES métier level 4 vocabulary.
 - The group question if this specific métier could not be already identified with the specific métier code “TBB_CRU_16-31_0_0” specifying especially specific mesh-size ranges regarded other métier as “TBB_DEF”. In this case, could be TBC only used as a national code?
 - The group argued that a new gear code could be added but should correspond to a real different gear that the ones already included in the list and not only to a specific target species or a specific mesh-size range. If added, the gear code should be added to the EU-MAP gear reference list also ICES vocabulary.
 - The group argued that the métier reference list at DCF level6 aims to provide harmonization/homogenization of codes between MS, métiers at DCF level7 could be used for further specification/national-regional needs. The group fears that adding such specific gear will open the gate for a huge variety of gears in the métier reference list which is not the aim of the métier reference list at DCF level6. National codes / DCF level7 should be preferred to better distinguish gears.
 - Finally, the group discussed the introduction of the gear code TBC in the reference lists with Torsten and **concluded that the brown shrimp fishery can be identified with the code TBB_CRU_16-31_0_0 and therefore not to introduce the TBC gear code in the reference lists.** The code should continue

⁴ Principles used for defining métier codes: I. Gear-target species assemblage combinations (métier level 5) follow table 5 from EU-MAP commission delegated decision (EU 2021/1167), ...

⁵ TBC: Trawl Beam/Bottom Crangon (i.e. brown shrimp, TBB_CRU_16-31). By value (one of) the largest single species fisheries in the North Sea. Has different impact on sea bottom - employs special gear (bobbin ropes) - than e.g. TBB_DEF>80 (tickler chains). Under discussion because of intensive use of Wadden Sea National Park areas (COM Action Plan; prohibition of mobile bottom contacting gears in marine protected areas until 2030).

⁶ Principles used for defining métier codes: I. Gear-target species assemblage combinations (métier level 5) follow table 5 from EU-MAP commission delegated decision (EU 2021/1167) ...



to be used for national purposes. It was noted that previously the code TBS has been used by NL, but that is not anymore possible with the new métier list. NL should use also TBB_CRU_16-31_0_0 for its brown shrimp fishery for seek of consistency.

- FSN stow net⁷.
 - The group highlight that this new fishing technique is rarely employed at this stage. May-be should be discussed further if this new fishing technique grow up?
 - The FSN gear code is in the EU-MAP commission delegated decision (EU 2021/1167) table 5 list as stow nets, and in the FAO list [Classification and illustrated definition of fishing gears \(fao.org\)](#) but from these definitions it seems that this gear code is more related to a pots/traps fishery which is not in line with the Germany fishery described in the request.
 - Finally, the group discussed the suggestion with Torsten and, since this gear is currently used by very few vessels in Germany and it doesn't seem that it is used by other countries, **concluded and decided not to include the FSN gear code yet in the métier reference lists**, but to keep aware that it might become a larger fishery in the future. If needed, the gear code to use should be discussed as FSN gear code detailed in FAO seems more related to a pots/traps fishery that the one described by Germany.

Request from RCG LP to include information on School type/FADs in the métiers reference lists (Mathieu from RCG LP joined the 2nd ISSG meeting as invited expert to discuss the request).

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- A short presentation was made by Mathieu regarding the importance of inclusion of information on School type (e.g. *Fishing Aggregating Devices, FADs*) for LP fisheries (*key element for these fisheries*).
- This information is currently used by France/Tropical purse seine fleets and is included in the FDI data call in the domain definition to provide quality biological information. Indeed, when estimating biological data, it is crucial to split fisheries regarding this information.
- This information is asked by the RFMOs ICCAT⁸ and IOTC⁹ for their specific data calls also included in WECAFC discussions on DCRF¹⁰. This information is then used for stock assessment and quotas evaluation.
- The group indicated that in RDBES CE table, “Number of FADs/Buoy” is a variable asked but the group noted that it should be associated to specific métiers considering only the ones which are performed on e.g. FADs.
- Artisanal fisheries in France/Other regions (*Réunion/Mayotte Islands in Indian Ocean; Guadeloupe/Martinique islands in West Indies*) used also FADs. For these fisheries, this information is crucial to estimate weight catch by trip (*LPUE*), species composition also for biological information (*fish size*).

⁷ FSN: stow net. Very different character of fishing technique and impact on the sea bottom than e.g. TBB. Rarely employed at the moment, but might become more interesting within the discussion of the COM Action Plan. Might be an alternative for the fishermen targeting brown shrimp (Crangon; TBB_CRU_16-31).

⁸ International Commission for the Conservation of Atlantic Tunas

⁹ Indian Ocean Tuna Commission

¹⁰ Western Central Atlantic Fishery Commission (WECAFC) – Data Collection Reference Framework (DCRF).

- The group discussed the issue and the way to introduce this information in the métier codification with Mathieu also if it should be added at DCF level6 or DCF level7 and **concluded and agreed to include this new information as optional information in the DCF level 7 métier codes reference list for LP fisheries.**
- Sébastien/Mathieu suggested codes to add in the DCF level7 métier reference list. Proposal is in accordance with how this information is asked in ORGPs CTOI/ICCAT and WECAFC data calls. To keep consistency in historical time series, the métier codes available nowadays will be kept as it is. It means that no information about FADS/School types means the information is unknown. Then the proposal is to give the possibility (optional) to MS to inform further in the métier code DCF level7 (after the group of species targeted) about the “FADs/School type” used distinguishing three different cases: “Fishing on schools associated with anchored floating objects”, “Fishing on schools associated with drifting floating objects” and “Fishing on free schools (unassociated with floating objects)”. These codes proposal have to be then evaluated by the ISSG and the RCG LP before being added in the JRC and ICES métier reference lists for data calls. At this stage the proposal is to add this possibility only for LP fisheries in RCG LDF area but RCG LP has to discuss the possibility to add them also in other RCGs’ LP fisheries.

Feedback from RCG LDF

- There was a comment from RCG LDF that they do not see the relevance of including métier at DCF level7 for their RCG and think that it could be confused¹¹.
- The group discussed the feedback and agreed to send a clarification to RCG LDF chair.
- A clarification was sent to the RCG LDF chair where it is clarified that metiers on DCF level7 have been, at this stage, only added to answer specific needs of the RCG LP and will not be asked for other fisheries especially the fisheries monitored by RCG LDF. ISSG understand the fear of possible confusion as RCG LDF and RCG LP could have fisheries in the same area but, as far as ISSG known the two RCGs have no responsibility for same fisheries. Therefore, we highlight that RCG LDF will not have to work on fisheries where metier DCF level7 will be asked and we agree not to include any specific DCF level7 métier for the fisheries monitored by RCG LDF.
- **The reply was accepted by the RCG LDF chair.**

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Issue with SX/SBV gear code

- **SX** gear codification (*combined beach and boat seines*) was added to the métier codes reference list for the cases where detailed gear is not available i.e. SB (*Beach seines*) or SV (*Boat or vessel seines*).
- This is coded as **SBV** in the ICES Vocabulary so confusing and not harmonized.

¹¹ The RCG was questioned about the added metiers on DCF level 7 which were established based on Spain's request for the Large Pelagic fisheries, as it was important for them to keep the target species for RCG LP. The RCG LDF considers that there is no need for metiers on level 7 to be assigned to this RCG even if some bycatch of large pelagic occurred in some areas. It was highlighted that the RCG LDF data-call requests submission of effort and landings data calculated on the metier level 6.

- The group noted that this mixed codification is not available in the EU-MAP commission delegated decision (EU 2021/1167) which is informed only for the gear codes SB (*Beach seines*) and SV (*Boat or vessel seines*).
- It is only Spain that has used the SX métiers in the STECF FDI data call, and only for the RCG MBS.
- In the FAO description, SX is described as something different i.e. a seine not included in the other seine category as SB or SV. <https://www.fao.org/fishery/en/geartype/204/en>. SBV is not a code available in FAO gear classification.
- **The group discussed the issue and agreed that it would be more correct to use the code SBV (i.e. SBV_DEF_>0_0_0 with the > added) for seek for consistency with ICES vocabulary and to better explain that it is a mixed code when SB or SV could not be detailed.**
- Spain was asked if they could update the full time series from “SX_DEF_0_0_0” to “SBV_DEF_>0_0_0” once the métier reference list will be updated. Spain asked that they could not update the full time series but can change the métier codes from 2023 onwards only if needed.
- **Following this answer, (and the fact that it is only Spain who has used the SX métiers in the FDI data call, and only for the RCG MBS), the group continued the discussions and agreed that it is better to accept the code as it is (SX_DEF_0_0_0) (and with a missing >) in order to keep the historical time series consistent.** Nevertheless, the group remembered that it would be more correct to use the code SBV (i.e. SBV_DEF_>0_0_0 with the > added) when possible to update the full time series to better explain that is a mixed code when SB or SV could not be detailed.
- ICES would be approached to see if it is possible for them to add SX gear (in addition to SBV) in their vocabulary (esp. for WGBYC data call).

Request from Azores for new métier code

Azores ask for the introduction of a new métier code GNS_MPD_>=100_0_0 (Set Gillnets, Mixed pelagic and demersal) at DCF level 6 in the métier reference list. The group discussed the proposal.

- Hugo (expert from Azores) explain that this is an important coastal fishery in Azores, at the shelf where they catch both pelagic and demersal fish. Hugo explain that in Azores this correspond to a real mixed fishery where both pelagic and demersal fish are targeted / caught without any preferences.
- The group noted that following métier codes: GNS_DEF, GNS_LPF & GNS_SPF with following mesh-size ranges “100-119_0_0”, “120-219_0_0” & “>=220_0_0” at DCF level6 already exists for RCG NATl
- Therefore, following the discussions, **the group concluded and agreed that the following métier codes could be added to the RCG NATl DCF level6 métier reference list to take into account the request and the specificity of this fishery: “GNS_MPD_100-119_0_0”, “GNS_MPD_120-219_0_0”, “GNS_MPD_>=220_0_0” and “GNS_MPD_>0_0_0” (for the cases where mesh-size range is not known).**
- The group noted that this mixed target group of species métier could not be inferred directly from the métier script, Azores should insert a national specificity in the R-script.

Correction of code

The métier codes **LA_LPF_<14_0_0_0**, **LA_SLP_<14_0_0_0** and **LA_SPF_<14_0_0_0** available in the DCF level6 métier reference list on the GitHub contained an extra zero, giving a warning in the script. They have been corrected to: **LA_LPF_<14_0_0**, **LA_SLP_<14_0_0** and **LA_SPF_<14_0_0** in the métier code reference list following the coding principles retained by the ISSG.

Request from Norway on métier codes used in FAO area 48

Norwegian catch statistics lists a large volume of catches in area 48 (Antarctic-Pacific) for which there are no appropriate métier codes in the métier reference list. For RDBES the valid codes in area 48 is listed hereunder:

OTB_MDD_70-119_0_0

OTM_DEF_70-119_0_0

The Norwegian catches in the area are krill, caught with either unknown trawl or pelagic trawl with mesh size 16 mm. Therefore, there is a need to add métier code with mesh size ranges adapted and targeting “crustaceans” i.e. “CRU”.

RCG ISSG suggestion:

- Area 48 is under LDF RCG.
- The RCG ISSG suggest to add the codes OTM_CRU_16-31_0_0 & OTM_CRU_>0_0_0 for the RCG LDF, to follow existing mesh size ranges and métier codification rules.
- The already existing code MIS_CRU_0_0_0 can be used when gear is unknown

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Task 2: Update métier descriptions from the 2023 RDBES data call (tables CE&CL). Consider the use of the métier description reports.

In 2023 a script for creating métier description reports based on RDBES effort and landings data (tables CE&CL) was developed. Reports are generated in a hierarchical way to describe the métiers on the different levels (4, 5 and 6). In 2023 data was available for one year, while in 2024 three years of data were available regarding the ICES RDBES data call.

The group highlights that first aim of such métier description will be to assist MS for defining métier, to gain in harmonization/homogenization between MS and also to support end-users when they have to combine/concatenate métiers for their needs.

A data extraction from RDBES CE and CL tables has been made available to the ISSG in beginning April (deadline RDBES data.call 2/04; 3 years of data 2021-2023 asked for tables CL and CE), and after some updates from MS made available again in early May.

Then the script developed last year was used to update the métier description html report. The script run annually, a task for 2024/2025 could be to adjust it to run multiannually. In the reports it is possible to see which countries and vessel lengths are using the métiers on the levels 4, 5 and 6. On métier levels 6, the landed weight and value of top 15 species are illustrated, and a map by ICES rectangle is showing where the fishery is taking place.

Following conclusions were made by the group:

Baltic Sea:

- OTM is ranked highest with both weight and value of landings in 2021, 2022 and 2023, and is used mainly by vessels larger than 18 m.
- GNS is ranked highest with fishing days in 2021, 2022 and 2023, and is used mainly by vessels less than 12 m.

North Sea and Eastern Arctic:

- OTM is ranked highest in landings weight in 2021, 2022 and 2023, and the highest landings are from vessels larger than 40 m.
- OTB is ranked highest in landings value in 2021, 2022 and 2023.
- In 2021 and 2022 TBB is ranked highest in fishing days, while it is OTB that is ranked highest in 2023.
- In 2021, 2022 and 2023 OTB is ranked highest in number of trips.

North Atlantic:

- OTM is ranked highest in landings weight in 2021, 2022 and 2023, and the highest landings are from vessels larger than 40 m.
- OTB is ranked highest in landings value in 2021, 2022 and 2023.
- In 2021 and 2023 OTB is ranked highest in fishing days, while FPO ranked highest in 2022.
- In 2021, 2022 and 2023 FPO is ranked highest in number of trips.

Example of overview on level 4 from 2023:

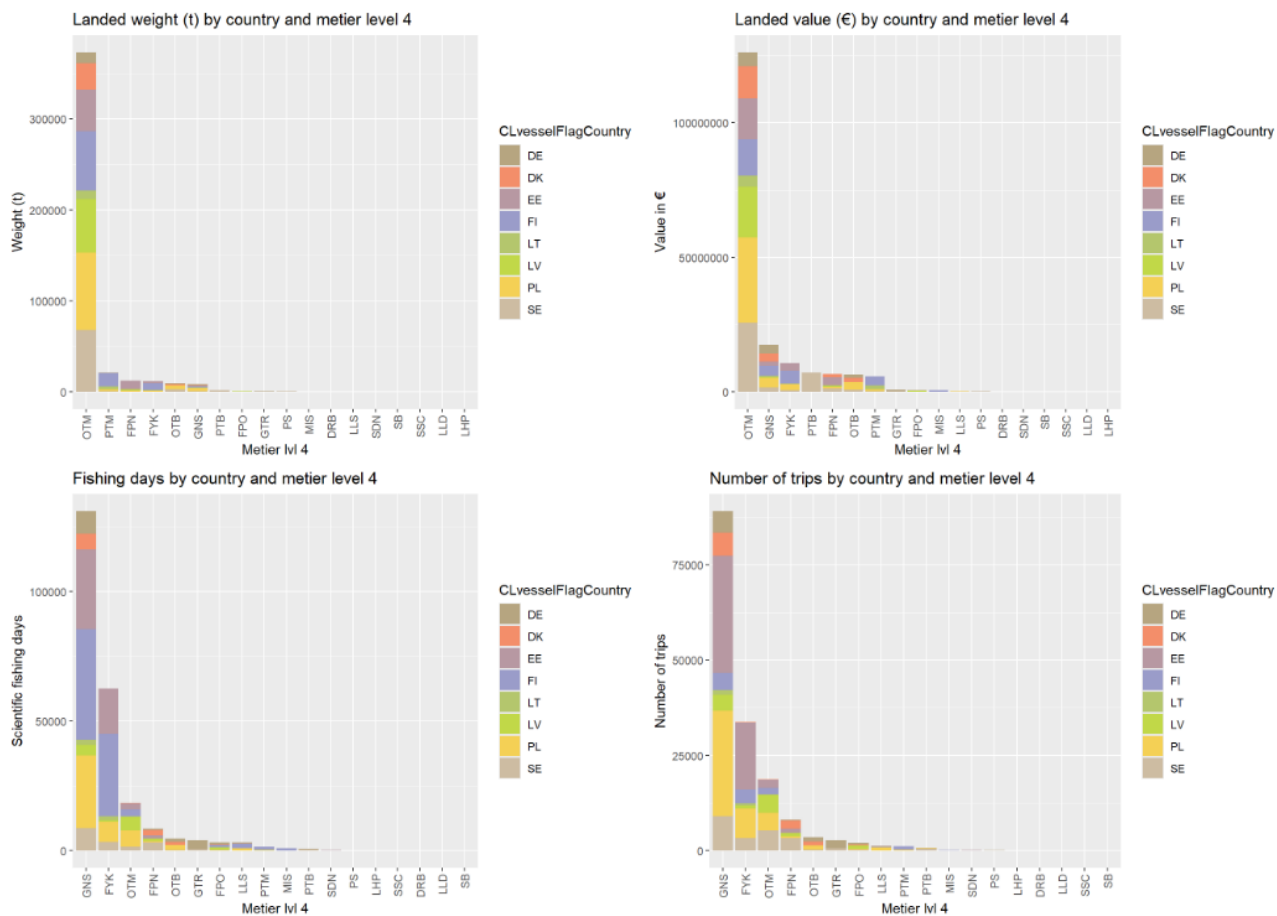


Figure I: example of overview on level 4 by vessel flag country from Baltic Sea 2023 metier description report, based on RDBES data.

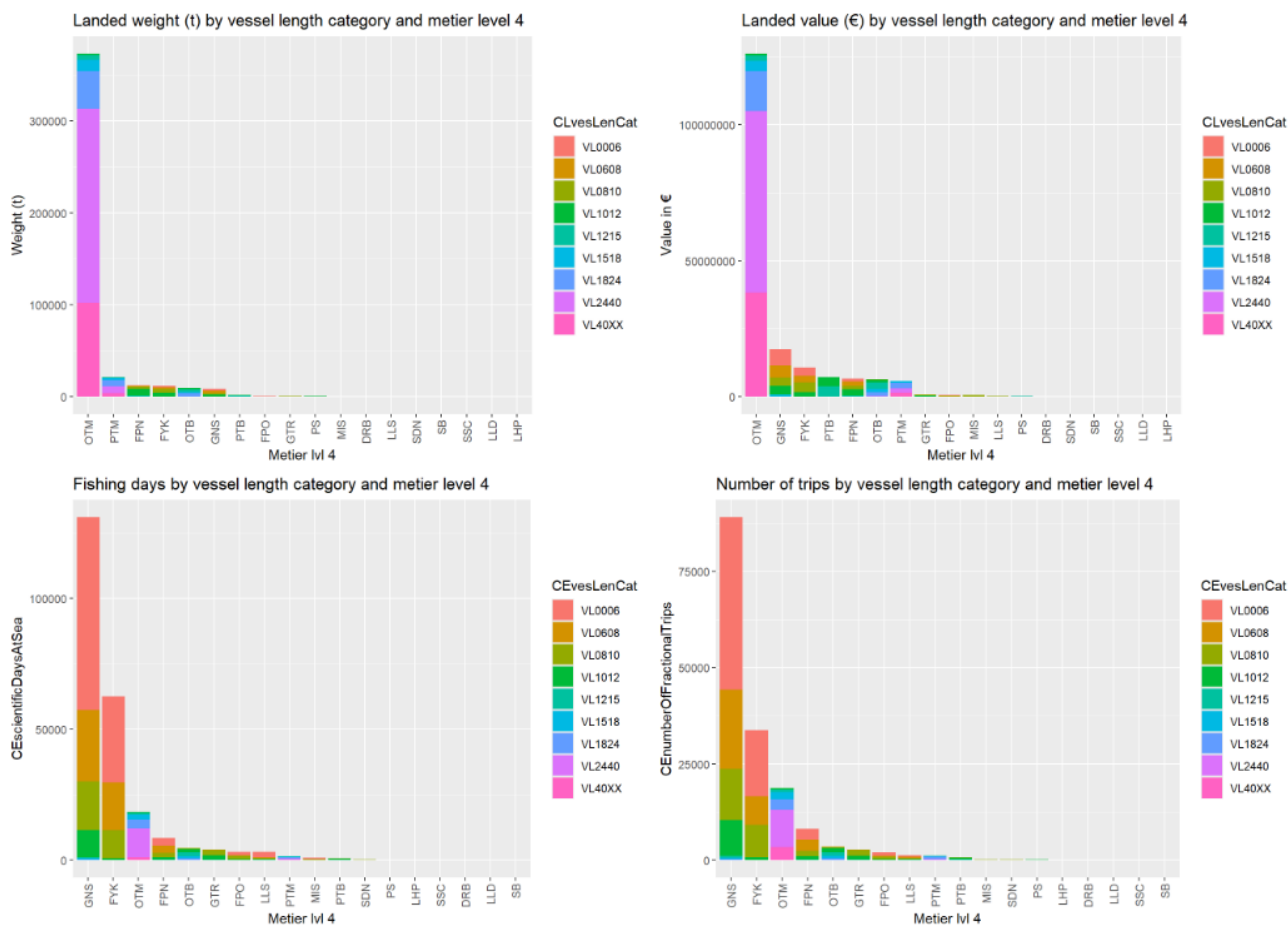


Figure 2: example of overview on level 4 by vessel length group from Baltic Sea 2023 métier description report, based on RDBES data.

The métiers were ranked by RCG region, and overviews were made of the top 10 métiers by weight of landings, value of landings and fishing effort measured as fishing days. The tables were made as a total for the years 2021-2023 (table 1-3) and for 2023 separately (table 4-6). By weight, it is the OTM_SPF métiers that is ranked highest in all 3 regions. By value, also métiers targeting SPF is ranked highest in the North Sea and Eastern Arctic and Baltic Sea regions, but in North Atlantic OTB_DEF ranked highest. By effort the picture is different with the TBB_CRU_16-31 ranked highest in North Sea and Eastern Arctic, GTN_DEF_>0_0_0 in North Atlantic and GNS_FWS_>0_0_0 in the Baltic.

Table 1: Top ten métiers ranked by total catch weight, for the period 2021 to 2023 across three regions: North Sea and Eastern Arctic, North Atlantic, and Baltic. Each métier's entry includes the weight in tonnes and the number of countries contributing to these metrics.

Regions	Métier level 6	Total weight (t)	Number of countries contributing
North Sea and Eastern Arctic	OTM_SPF_32-69_0_0	977067	7
	OTB_DEF_<16_0_0	301704	4
	OTM_SPF_16-31_0_0	246013	5
	DRB_MOL_>0_0_0	183419	5
	OTB_DEF_>=120_0_0	125336	9
	TBB_CRU_16-31_0_0	73452	4

North Atlantic	HMD_MOL_>0_0_0	57990	1
	OTM_DEF_16-31_0_0	53926	3
	PTM_SPF_32-69_0_0	45687	4
	OTB_CRU_32-69_2_22	37743	2
	OTM_SPF_40-54_0_0	558893	5
	PTM_SPF_40-54_0_0	318594	2
	OTM_DEF_40-54_0_0	174984	2
	PS_SPF_>0_0_0	167928	2
	HMS_SWD_0_0_0	166164	1
Baltic	OTB_DEF_100-119_0_0	97629	3
	DRB_MOL_>0_0_0	78728	3
	OTM_DEF_32-39_0_0	71483	2
	OTM_SPF_32-39_0_0	51409	3
	GNS_DEF_100-119_0_0	47026	3
	OTM_SPF_16-31_0_0	911864	7
	OTM_SPF_>0_0_0	201368	1
	PTM_SPF_16-31_0_0	74439	6
	OTM_SPF_32-89_0_0	63721	7
	FPN_SPF_>0_0_0	35981	6
	FYK_FWS_>0_0_0	18986	7
	OTB_DEF_>=120_3_120	17461	3
	GNS_FWS_>0_0_0	13462	8
	FYK_SPF_>0_0_0	12859	5
	GNS_DEF_110-156_0_0	7981	8

Table 2: Top ten métiers ranked by total economic value, for the period 2021 to 2023 across three regions: North Sea and Eastern Arctic, North Atlantic, and Baltic. Each métier's entry includes the economic value in euros and the number of countries contributing to these metrics.

Regions	Métier level 6	Total value (€)	Number of countries contributing
North Sea and Eastern Arctic	OTM_SPF_32-69_0_0	638107384	7
	TBB_CRU_16-31_0_0	367392385	4
	OTB_DEF_>=120_0_0	323121321	9
	DRB_MOL_>0_0_0	312083986	5
	TBB_DEF_70-89_0_0	190360401	3
	TBB_DEF_70-99_0_0	153097867	3
	OTB_CRU_32-69_0_0	126375212	5
	OTB_DWS_>=120_0_0	112303450	2
	OTM_SPF_16-31_0_0	107705341	5
	OTB_DEF_<16_0_0	93829535	4
North Atlantic	OTB_DEF_100-119_0_0	348998166	3
	OTM_SPF_40-54_0_0	202133680	5
	DRB_MOL_>0_0_0	195255165	3
	PTM_SPF_40-54_0_0	192325103	2
	FPO_MOL_>0_0_0	184580856	3
	GNS_DEF_100-119_0_0	161261560	3
	PS_SPF_>0_0_0	147950378	2
	OTB_CRU_70-99_0_0	146574908	3

Baltic	TBB_DEF_70-99_0_0	142133157	4
	LLS_DEF_0_0_0	141149017	2
	OTM_SPF_16-31_0_0	259745607	7
	OTM_SPF_>0_0_0	41779322	1
	GNS_FWS_>0_0_0	27867568	8
	OTM_SPF_32-89_0_0	21702367	7
	FYK_FWS_>0_0_0	21041967	7
	PTB_FWS_>0_0_0	20341398	2
	PTM_SPF_16-31_0_0	17755374	6
	GNS_DEF_110-156_0_0	12590675	8
	FPN_SPF_>0_0_0	10609565	6
	OTB_DEF_>=120_3_120	10371399	3

Table 3: Top ten métiers ranked by total fishing effort, for the period 2021 to 2023 across three regions: North Sea and Eastern Arctic, North Atlantic, and Baltic. Each métier's entry includes the effort measured in fishing days and the number of countries contributing to these metrics.

Regions	Métier level 6	Total effort	Number of countries contributing
North Sea and Eastern Arctic	TBB_CRU_16-31_0_0	143811	4
	DRB_MOL_>0_0_0	71073	5
	FPO_CRU_>0_0_0	58831	7
	GNS_DEF_120-219_0_0	26039	6
	OTB_DEF_>=120_0_0	22912	9
	OTB_MCD_90-99_1_300	22297	2
	OTB_DEF_70-89_0_0	21875	4
	TBB_DEF_70-99_0_0	21617	3
	TBB_DEF_70-89_0_0	21265	3
North Atlantic	FPO_MOL_>0_0_0	20441	5
	GTN_DEF_>0_0_0	152387	2
	FPO_MOL_>0_0_0	129291	3
	GTR_DEF_>0_0_0	90476	2
	DRB_MOL_>0_0_0	88362	3
	FPO_CRU_>0_0_0	82357	3
	LLS_DEF_0_0_0	70313	2
	DRH_MOL_>0_0_0	65094	1
	OTB_DEF_100-119_0_0	64578	3
Baltic	GTR_DEF_100-119_0_0	59345	2
	OTT_CRU_70-99_0_0	46197	1
	GNS_FWS_>0_0_0	259951	8
	FYK_FWS_>0_0_0	126007	7
	GNS_DEF_110-156_0_0	88646	8
	OTM_SPF_16-31_0_0	41822	7
	FYK_ANA_>0_0_0	36602	4
	GNS_SPF_32-89_0_0	35432	7
	FYK_SPF_>0_0_0	16586	5
	GNS_ANA_>=157_0_0	14356	7
	GNS_DEF_>=157_0_0	13887	6
	FYK_CAT_>0_0_0	11220	2

2023

Table 4: Top ten métiers ranked by total catch weight, for the year 2023 across three regions: North Sea and Eastern Arctic, North Atlantic, and Baltic. Each métier's entry includes the weight in tonnes and the number of countries contributing to these metrics.

Regions	Métier level 6	Total weight (t)	Number of countries contributing
North Sea and Eastern Arctic	OTM_SPF_32-69_0_0	347321	7
	OTB_DEF_<16_0_0	129526	3
	OTM_SPF_16-31_0_0	103531	4
	DRB_MOL_>0_0_0	54491	5
	OTB_DEF_>=120_0_0	50363	8
	OTM_DEF_<16_0_0	26678	3
	TBB_CRU_16-31_0_0	21148	4
	OTB_CRU_32-69_2_22	13605	2
	OTB_CRU_32-69_0_0	13151	5
	HMD_MOL_>0_0_0	12803	1
North Atlantic	OTM_SPF_40-54_0_0	206081	5
	PTM_SPF_40-54_0_0	100614	1
	OTM_DEF_40-54_0_0	93422	2
	PS_SPF_>0_0_0	57015	2
	HMS_SWD_0_0_0	49601	1
	OTM_DEF_32-39_0_0	33317	2
	OTB_DEF_100-119_0_0	31790	3
	DRB_MOL_>0_0_0	26990	3
	OTM_SPF_32-39_0_0	26264	2
	GNS_DEF_100-119_0_0	14938	2
Baltic	OTM_SPF_16-31_0_0	289165	7
	OTM_SPF_>0_0_0	65480	1
	PTM_SPF_16-31_0_0	20761	5
	OTM_SPF_32-89_0_0	17946	4
	FPN_SPF_>0_0_0	11786	6
	FYK_FWS_>0_0_0	6205	7
	FYK_SPF_>0_0_0	5054	5
	OTB_DEF_>=120_3_120	4513	3
	GNS_FWS_>0_0_0	4202	8
	OTB_SPF_16-31_0_0	2826	4

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Table 5: Top ten métiers ranked by total economic value, for the year 2023 across three regions: North Sea and Eastern Arctic, North Atlantic, and Baltic. Each métier's entry includes the economic value in euros and the number of countries contributing to these metrics.

Regions	Métier level 6	Total value (€)	Number of countries contributing
North Sea and Eastern Arctic	OTM_SPF_32-69_0_0	239662357	7
	OTB_DEF_>=120_0_0	131464237	8
	TBB_CRU_16-31_0_0	124883622	4

North Atlantic	DRB_MOL_>0_0_0	114173297	5
	TBB_DEF_70-89_0_0	85687132	3
	OTM_SPF_16-31_0_0	55589755	4
	OTB_CRU_32-69_2_22	54387956	2
	OTB_DWS_>=120_0_0	53792265	2
	OTB_CRU_32-69_0_0	47725077	5
	OTB_DEF_<16_0_0	40067310	3
	OTB_DEF_100-119_0_0	129449110	3
	OTM_SPF_40-54_0_0	73134849	5
Baltic	DRB_MOL_>0_0_0	68108415	3
	PTM_SPF_40-54_0_0	65333348	1
	GNS_DEF_100-119_0_0	55063480	2
	FPO_MOL_>0_0_0	54375922	3
	PS_SPF_>0_0_0	52789469	2
	TBB_DEF_70-99_0_0	49741974	3
	OTB_CRU_70-99_0_0	47318495	3
	GTR_DEF_100-119_0_0	46968136	2
	OTM_SPF_16-31_0_0	104081919	7
	OTM_SPF_>0_0_0	13538757	1
	GNS_FWS_>0_0_0	8780323	8
	OTM_SPF_32-89_0_0	8264771	4
	FYK_FWS_>0_0_0	7260213	7
	PTB_FWS_>0_0_0	7065170	2
	PTM_SPF_16-31_0_0	5708321	5
	FPN_SPF_>0_0_0	3991173	6
	GNS_DEF_110-156_0_0	3892316	8
	OTB_DEF_>=120_3_120	2962429	3

Table 6: Top ten métiers ranked by total fishing effort, for the year 2023 across three regions: North Sea and Eastern Arctic, North Atlantic, and Baltic. Each métier's entry includes the effort measured in fishing days and the number of countries contributing to these metrics.

Regions	Métier level 6	Total effort	Number of countries contributing
North Sea and Eastern Arctic	TBB_CRU_16-31_0_0	39815	4
	DRB_MOL_>0_0_0	22796	5
	FPO_CRU_>0_0_0	18643	7
	TBB_DEF_70-89_0_0	8720	3
	GNS_DEF_120-219_0_0	8204	4
	OTB_DEF_>=120_0_0	6732	8
	OTB_CEP_70-89_0_0	6717	2
	OTB_MCD_90-99_1_300	6596	2
	FPO_MOL_>0_0_0	6236	3
	OTB_DEF_70-89_0_0	5910	4
North Atlantic	GTN_DEF_>0_0_0	49246	2
	FPO_MOL_>0_0_0	38262	3
	GTR_DEF_>0_0_0	34871	2
	DRB_MOL_>0_0_0	29156	3
	LLS_DEF_0_0_0	28162	2

Baltic	FPO_CRU_>0_0_0	24435	3
	OTB_DEF_100-119_0_0	19878	3
	GTR_DEF_100-119_0_0	18951	2
	DRH_MOL_>0_0_0	14951	1
	OTT_CRU_70-99_0_0	13750	1
	GNS_FWS_>0_0_0	75980	8
	FYK_FWS_>0_0_0	42229	7
	GNS_DEF_110-156_0_0	27350	8
	OTM_SPF_16-31_0_0	13669	7
	GNS_SPF_32-89_0_0	11930	7
	FYK_ANA_>0_0_0	11883	3
	FYK_SPF_>0_0_0	5281	5
	GNS_ANA_>=157_0_0	4957	6
	GNS_ANA_110-156_0_0	3573	6
	GTR_DEF_110-156_0_0	3479	2

The métier description reports will be made available on the RCG sharepoint and accessible for experts who have access to it.

Future work:

- Create a multiannual métier report, with averages over the years, and a graph showing the annual variation, in addition to an annual report for the last year.
- When the RDBES data license/confidentialities issues has been updated, develop (in addition to the internal version) a public version of the report where, for example, all figures with less than 2 vessels are removed i.e. show information only where there are more than 2 vessels (3 vessels at least). Such rules should be evaluated to see how it will impact the report, what will be missing.

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Task 3: Questionnaire on cross-validation methods used by MS to combine data from different sources

This task work has been based on the replies from last year questionnaire (sent from the ISSG to NCs in spring 2023) regarding methodologies used by MS to calculate fishing activity estimates. The questionnaire evaluates especially the use of cross-validation methods in MS to combine data coming from different declarative sources.

As the replies to the questionnaire were both very comprehensive and diverse, they were first summarized in the ISSG Métier and transversal variables issues last year' report. This year, work has been devoted to a selection of questions in order to identify common/standard practices used in MS and to highlight practices which could have the potentiality to enhance data quality. The focus this year has been on the three following topics:

1. What is the type of declarative data considered to calculate/assess fishing activity data?
2. The algorithms/methods used to cross-validate/combine the different types of declarative data available to consolidate/optimize national fishing activity data (e.g. bringing together fishermen logbooks and sales notes data).
3. The algorithms/methods used to estimate/assess the value of landings (esp. for landings not sold at auctions).

Overview of conclusions for these different topics are summarized hereunder.

Topics	Description	Potential to enhance data quality
Type of declarative data considered to calculate/assess fishing activity data.	Logbooks and Coastal logbooks, principal data sources. Sales note and geo-location data, major other data sources.	Following the implementation of obligation on keeping an electronic fishing logbook by all Union vessels (<i>deadline 2028</i>) including <10m SSF vessels, develop coastal e-logbooks adapted to their specificities to monitor SSF fishing activity. Consider e.g. a sales event and/or a VMS position also when not assigned to a declared fishing trip to improve data completeness and develop concomitantly SSF geo-localization data. Consider sales note and geo-location data to be cross-checked with declared fishing trips. Consider complementary data (e.g. <i>fishing licenses</i>) to further improve data quality.
Algorithms/methods to cross-validate/combine the different types of declarative data available to consolidate/optimize national fishing activity data.	Two major approaches: 1) Same identifiers shared 2) Algorithm developed based first on vessel & dates and eventual other information (e.g. <i>species composition</i>)	Develop use of a common identifier and share MS' algorithms to improve data combination. Avoid using unique data source. Encourage implementation of cross-validation methods to enhance data quality and completeness.
Algorithms/methods to estimate/assess the value of landings (esp. for landings not sold at auctions).	Three different approaches: 1) Directly issued from sales note data considered as complete 2) "Global" average-based approach derived from a general average price from sales note data 3) Specific algorithm combining linked sales note data if available and calculated average price when needed	Encourage developing specific algorithm combining information directly deducted from linked sales notes data if available and calculated average price when needed. Consider spatial (e.g. <i>landing harbor</i>), temporal (e.g. <i>month</i>) and vessel characteristics (e.g. <i>vessel length ranges</i>) information to calculate average price per species when needed.

		Use of expert knowledge or additional information could be needed to complete information.
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More details can be found in annex 2, where for each question, the first paragraph “Summarized information” presents information detailed in the last year’ report to introduce new readers and in the second paragraph some of the practices are discussed more in depth especially regarding their potential to enhance data quality. Finally, the replies by country related to the topic have been summarized and are presented in the appendix of the annex (*one appendix by topic*).

The other topics/questions will be included in next year’s work of the group. The final aim of this task, in intermediate term, would be to suggest and develop best practices guidelines for cross-validation methods to calculate/improve fishing activity data with a specific focus on the RDBES fishing activity data (i.e. variables asked in the RDBES tables CE&CL) and an objective of enhancing fishing activity data quality and completeness. It will take the form of a stand-alone document compiling all the conclusions done during the ISSG years’ work on this issue.

Task 4: Work on a template to document CE and CL data uploaded to RDBES (documentation of methods)

In 2023, it was considered that ISSG is now dealing with transversal variables in general which are especially reported in the RDBES CE and CL tables. Based on questionnaire information, the group is working on best practices guidelines, and considers useful to have a common format to document the transversal data (effort and landings) that are submitted to the RDBES (i.e. having a good and comprehensive documentation of the methodologies used to calculate fishing activity estimates for data calls). Such documentation will be essential to enhance harmonization between MS and will be a necessary first step in the move towards the development of best practice guidelines that ultimately consider the large diversity of data sources and methodologies being used (e.g., with regards to SSF).

A template to document methods used for processing the CE and CL data uploaded to the RDBES have been elaborated and is available in Annex I¹². Part of the information can be retrieved directly from the data or ‘upload logs’, but the ISSG considers that here more details can be given and that it is useful to have a common descriptive format that can also be publicly shared with users of data that doesn’t have access to data and upload logs.

The template was created based on the STECF FDI national chapters, FDI methodology meeting report, AER/FDI workshop report and AER/FDI questionnaire developed in 2023 by the ISSG.

The ISSG suggests that the usefulness of documenting the data in this format should be discussed at the RCG NANSEA & Baltic TM in June 2024 in the planned subgroup on RDBES.

Task 5: FecR package

¹² The group discussed the eventual overlap of such task with the new WGRDBESGOV Quality ISSG subgroup and agreed that ISSG task will focus on the methods when Quality ISSG will focus on data.

The aim of this task is to continue to follow up the development of the fecR package and its effort to calculate fishing effort metrics that are harmonized/homogenized between MS. Specific focus should be done, in collaboration with ISSG SSF, to include SSF in fecR calculation.

The fecR package provides a set of functions that implement the so called “Nicosia principles for fishing effort calculation” with the aim to standardize the calculation of the fishing effort metrics “fishing days” and “days at sea” by MS to answer data calls (e.g. STECF FDI data call). The development of fecR started during the 2nd Workshop on Transversal Variables (22-26 February 2016) and a first version was put online in a public repository (CRAN) in early November 2016. The use of the package for MS effort calculations was then promoted in the 2017 and 2018 FDI data calls but in December 2018 the package was put offline and archived by CRAN after its code failed to pass a few internal checks. CRAN registered difficulties in contacting the maintainer of the package. From that moment to present, the package remained offline with only archived versions being available to MS a situation that complicated its usage in the answering of data calls. In 2023 the ISSG agreed not to have the ambition to put the package on CRAN again, but to make it available in a public repository. From that, the fecR package R scripts have been made available, since last year, through a GitLab hosted at the JRC: <https://dcallnet.jrc.ec.europa.eu/gitlab/transversal/fecR> (which requires an approved EU login). The package can be found also at <https://rdr.io/cran/fecR/>. The ISSG suggest that the updated code should be publicly available, and not requiring an EU login.

Through this task, the group has the ambition to continue to help and assist fecR package development to enhance harmonization/homogenization between MS when calculating fishing effort metrics. Furthermore, the group considers that following improvements could be made of the fecR package:

- Updated the package to the newest version of R.
- Make the script suitable for the RDBES aggregation.
- Consider that increasingly the logbook data are reported on haul level.
- Handling of small-scale fishery, considering their specificities also regarding specific type of data sources available.
- Check the FecR package against the new control regulation and minimum logbook requirements which will be asked in it.

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The group agreed to postpone most of the tasks of this work to next years as there was already a lot of topics to deal with by the ISSG and as most competent, for this task, group members were not available this year. The unique ambition of the group for this year work was to see if the R package can be updated to the newest version of R and doing that trained some more members to the fecR package to be able to help to improve it in the future.

The first conclusions after checking the operability of the fecR package were favorable with regard to the correct performance of the main functions. Following the decision not to include it in CRAN, the package does not need to pass any internal checks and calculates the fishing effort without any version issues. A grammar issue within the code was addressed related to the fishing areas list format. A publicly available code can be downloaded from the GitLab repository (<http://dcallnet.jrc.ec.europa.eu/gitlab/r-packages/fecR.git>).

Task 6: RCG Econ alternative fleet segmentation' initiative

The aim of this task for the group is to continue to follow up the development under RCG Econ of an alternative fleet segmentation and advice on it in order to keep the link between the two approaches (*métiers and fleet segmentation*). The work to do on this task by the group was fully dependent of the feedbacks of the

3rd workshop (held in May 2023) on this topic which has considered also some development made by the ISSG in 2023.

The ISSG chairs were in contact with the chairs of the RCG Econ fleet segmentations workshops, and got following feedback regarding the ongoing process and the expected workplan:

- No workshop or any coordinated action is planned at the moment, but the alternative approach is steadily developing.
- The method has been tested in Iceland, to characterize their demersal trawl fleet and in Romania.
- The approach is tested on the German fleet with an AI-extension.

As there was no plan developed at this time on how to proceed to advance on this task from RCG Econ fleet segmentations workshops chairs, as there was no task expected from ISSG to advance it and as there was no work/tasks listed for ISSG (expected from the ISSG) in the RCG Econ report and Alternative fleet segmentation workshop reports (which was checked by the ISSG), no further work on the approach was done from the ISSG side but the issue was kept in its roadmap for the next years.

It should be discussed at the Liaison meeting in September if the ISSG should continue following the RCG Econ fleet segmentation and the tasks on which the ISSG could have some input. Anyway, the two approaches are linked and a new alternative fleet segmentation should take into consideration the different métiers practiced by the vessels during the year (especially for the pre-segmentation step discussed during the last workshop). For example, such alternative fleet segmentation could consist in bringing together in fleet segments, vessels having relatively homogeneous annual exploitation strategies of métiers.

New Control Regulation and revision of the Implementation Act

The new control regulation (EU 2023/2842) has been agreed, and can be found here: https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ:L_202302842

The ISSG has been informed that the main features of the new control regulation are:

1. Full digitalisation with tracking and catch reporting for all fishing vessels
2. For small scale fisheries, there will be electronic catch reporting (also fishing without a vessel) and position data
3. For recreational fisheries there will be electronic catch reporting and data collection
4. The rules regarding data access and protection will be clearer

The ISSG was also informed that over the next two years, the existing implementing act (EU 404/2011) (<https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX%3A32011R0404> ; where all the details for the implementation of the regulation are explained) will be reviewed (which e.g., details logbook reporting). DG-MARE encouraged the group to make review from the data collection (RCG) side on this, e.g., on effort variables reporting and to provide feedbacks.

As a first comment, the group highlight that, currently, there is a mis-match in transversal variables listed in table 6 of the EU-MAP (EU 2021/1167) that MS should be able to report and what is optional in the current Implementation Act of the control regulation (difference between control regulation – what is mandatory and scientific/DCF regulation). The ISSG recommends that the variables that are mandatory to report in the EU-

MAP table 6 are also mandatory in the new Control Regulation Implementation Act, focusing on annexes X, XI and XII.

Examples of variables that are listed in the EU-MAP table 6 not mandatory in Control Regulation Implementation Act. Currently the

Variables in EU MAP (EU 2021/1167) table 6	Control Regulation Implementation Act (EU 404/2011)	Comment
Days at sea	Annex X Day, month, hour (local) and port of departure and return (M)	OK
Hours fished (optional)	Fishing time (O). Total time spent searching (e.g. using sonar) or fishing shall be given and equals the number of hours spent at sea minus the time spent in transit to, between and returning from the fishing grounds, dodging, inactive or waiting for repair.	In DCF hours fished would normally be interpreted as only fishing, while the fishing time in the Control Regulation involves searching.
Fishing days	Date (M). Date for each day at sea shall be recorded in a new line and shall correspond to each day at sea.	OK
kW * Days at sea	Annex X Day, month, hour (local) and port of departure and return (M). kW is recorded in fleet register.	OK
GT * Days at sea	Annex X Day, month, hour (local) and port of departure and return (M). GT is recorded in fleet register.	OK
kW * Fishing days	Date (M). Date for each day at sea shall be recorded in a new line and shall correspond to each day at sea. kW is recorded in fleet register.	OK
GT * Fishing days	Date (M). Date for each day at sea shall be recorded in a new line and shall correspond to each day at sea. GT is recorded in fleet register.	OK
Number of trips	Annex X: Logbooks are filled by fishing vessels and trip dates.	OK
Number of fishing operations	Annex X: Number of fishing operations (M)	OK
Length of nets (m) * soak time (days)	Annex X Gear dimensions (O). Gear size and dimensions shall be given according to the specifications in column 2 of Annex XI. Annex XI for gillnets and entangling nets: Length, height	Optional in control regulation
Number of nets/length	Annex X Gear dimensions (O). Gear size and dimensions shall be given according to the specifications in column 2 of Annex XI. Annex XI for gillnets and entangling nets: Length, height	Optional in control regulation
Number of hooks, number of lines	Annex X Gear dimensions (O). Gear size and dimensions shall be given according to the specifications in column 2 of Annex XI. Annex XI for hooks and lines: Total number of hooks/lines shot each day	Optional in control regulation

Number of pots, traps	Annex X Gear dimensions (O). Gear size and dimensions shall be given according to the specifications in column 2 of Annex XI. Annex XI for traps: Number of pots shot each day	Optional in control regulation
Number of FADs/boys		
Number of support vessels		

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Annex I: Template to document methods applied to prepare commercial landings (CL) and effort (CE) data submitted to RDBES.

This document aims to collect information on the preparation of data submitted in response to the joint RCG-ICES data call. Some of this information is already contained in specific fields of RDBES data format or in the upload logs. However, this can only give a general overview of the methodologies used, while this document asks to provide more detailed information.

- 1. Describe data origin for scientific values of transversal variables. This can be either official or estimated. Official data can be based either on census or official sampling. If official sampling is in place, please provide a brief description or a reference to the documentation. If values are estimated, please describe methods used for estimation or re-evaluation. In case when different methodology are applied to different fleet segments, please specify.**

Group variables	of	Variable name	Methodology applied
Landings		Scientific weight	
		Landings value	
Vessel effort		Scientific days at sea	
		Scientific kW Days at sea	
		GT Days at sea	
		Number of fractional trips	
		Number of dominant trips	
		Number of unique vessels	
Gear effort		Scientific fishing days	
		Scientific kW Fishing days	
		GT Fishing days	
		Scientific vessel fishing hour	
		Scientific kW fishing hours	
		GT Fishing hours	
		Scientific soaking meter hour	
		Scientific number of hauls/sets	
		Gear dimensions	

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- 2. Describe data sources for scientific values of transversal variables. The most common are logbooks, sales notes, other declarative forms, sampling data, etc. Please provide a brief description of data sources and indicate if they vary according to vessel length categories (small-scale fleet vs large scale fleet), areas or other factors.**

Group variables	of	Variable name	Data sources
Landings		Scientific weight	
		Landings value	
Vessel effort		Scientific days at sea	
		Scientific kW Days at sea	
		GT Days at sea	
		Number of fractional trips	

Gear effort	Number of dominant trips	
	Number of unique vessels	
	Scientific fishing days	
	Scientific kW Fishing days	
	GT Fishing days	
	Scientific vessel fishing hour	
	Scientific kW fishing hours	
	GT Fishing hours	
	Scientific soaking meter hour	
	Scientific number of hauls/sets	
	Gear dimensions	

3. Scientific vs official landings weight.

Were differences between official and scientific landings weights identified? If yes, please provide a brief explanation of the differences and describe the calculations that caused the differences

4. Provide information on data completeness and availability.

Were all data on fishing activities available for responding to the RDBES data call? If not, specify the missing data.

Has information on all species landed been submitted to the RDBES? If not, please indicate which were not submitted.

Has information from all vessels been submitted to the RDBES? Is there any part of the fleet that has not been included in the data submitted?

5. Describe methods used to determine métier level 6 codes.

Do you use the script developed by the ISSG Transversal issues? Does the script require national adaptations or additions? If you do not use the script by ISSG, do you use similar algorithms to assign métier codes? If you do not use similar algorithms, specify.

Does your source data contain all the information needed to assign métier codes? If not, specify.

Is the target assemblage determined post factum using dominance criteria as in the script by ISSG or is it known from declarative forms?

How do you deal with a situation when a métier code could not be assigned?

Do you use vessel patterns to reassign rare métiers? If yes, please specify the thresholds applied.

Other information.

6. Describe methods used to calculate effort variables.

Is fishing effort calculated according to the principles specified at the 2nd Workshop on Transversal Variables, Nicosia, 2016?

Do you use the fecR package to calculate fishing effort? If not, do you use similar algorithms to those implemented in the fecR package? If you do not use similar algorithms, specify how you distribute effort to areas and gears?

Does your source data contain all the information needed to calculate effort of small scale fleet? If not, specify .

7. Describe methods used to provide spatial information on landings and effort.

What is a source of information on statistical rectangle? Is it registered in logbooks or other declarative forms? If it is estimated, please provide description of methods.

How is the exclusive economic zone determined?

Do you validate spatial information (statistical rectangles, FAO areas, EEZ) using other data sources, e.g. VMS, AIS, etc.?

Is the spatial information from small scale fleet available in declarative forms? If not, are there any geo-localisation systems used to identify location for small scale fleet activity?

8. Provide information on FDI related fields.

In the data model, there are fields required by FDI database, i.e. 'Fishing Technique', 'Mesh size range', 'Supra Region', 'Geographical indicator', 'Specific conditions to technical measures', 'Exclusive Economic Zone (EEZ)', 'Exclusive Economic Zone Indicator (EEZI)', 'Deep Sea Regulation', 'Official vessel hours at sea', 'FDI confidentiality code'. Do you provide data in all FDI related fields in the data submitted to the RDBES? Please briefly describe the methodology used to complete the FDI fields.

Are values of the relevant FDI fields (Supra Region, Geographical indicator, Fishing Technique) aligned with the values provided to the AER data call?

Annex 2: Questionnaire on cross validation methods used by MS to combine data from different sources

In 2023 a questionnaire regarding methodologies used to calculate fishing activity estimates was sent out to MS. It evaluates especially the use of cross-validation methods in MS to combine data coming from different declarative sources.

As the replies to the questionnaire were both very comprehensive and diverse they were first summarized in the ISSG Metier and transversal variables issues last year' report. This year, work has been devoted to a selection of questions in order to identify common/standard practices used in MS and to highlight practices which could have the potentiality to enhance data quality.

The focus this year has been on the three following topics:

- What is the type of declarative data considered to calculate/assess fishing activity data?
- The algorithms/methods used to cross-validate/combine the different types of declarative data available to consolidate/optimize national fishing activity data (e.g. *bringing together fishermen logbooks and sales notes data*).
The algorithms/methods used to estimate/assess the value of landings (*esp. for landings not sold at auctions*).

Below, for each question, the first paragraph “Summarized information” presents information detailed in the last year' report to introduce new readers, and in the second paragraph some of the practices are discussed more in depth especially regarding their potential to enhance data quality. Finally, the replies by country related to the topic have been summarized and are presented in the appendix (one appendix by topic).

Type of declarative data considered to calculate/assess fishing activity data.

1. Summarized information

All MS reported the use of the logbooks for vessels over 10 m for effort data. For effort calculation for vessels below 10 m monthly declarative form is used by 3 countries, sales notes are used by 2 countries, logbooks/monthly journals are used by 6 countries and 1 country has not specified the sources. However, in some cases sources are available for part of the fleet. Additional resources are: self-sampling program, observers at sea program and port sampling program which have been used by one MS. One MS raised the issue to report and calculate the fishing from ice. Questionnaire for sales data is used by one country, other obtained data from sales notes. When MS provides information on the geo location data, they reported the VMS data use as required in the Control Regulation. AIS data is collected only by one country. In general, data on SSF vessel position is missing.

2. Description of practices and discussion on potential to enhance data quality

Logbooks ($\geq 10\text{m}/8\text{m}$ in Baltic) or coastal logbooks ($< 10\text{m}/8\text{m}$ in the Baltic) constitute predominantly the principal (sometimes the unique) data sources considered to calculate/assess fishing activity estimates (eventually completed with landing declarations). In the majority of cases, a fishing trip will not be considered if no logbook/coastal logbook is available. Thus, the potential to complete fishing activity data by considering other data sources is little considered. E.g. a sales event and/or a VMS position (estimated with the “fishing” status) not assigned to a declarative fishing trip could constitute an evidence of a missing declaration. Developing such possibility is viewed as a way to improve data completeness.

Coastal logbooks (named differently in each country e.g.: monthly catch reports, monthly declarative fishing forms, coastal monthly logbooks, monthly journals, coastal logbooks) have been developed in several countries to monitor $< 10\text{m}$ SSF vessels’ fishing activity. Providing similar information as European logbooks, they present the advantage to be better adapted to the SSF specificities but, in the same time, are often less comprehensive (e.g. reporting is done by month and not by fishing trip). Nevertheless, development of such adapted declarative forms constitutes a way to improve data quality in particular for SSF fisheries. This is especially the case when SSF only have been monitored via sales notes data. Sales notes data provides insufficient information to meet end-users’ needs even when (but it is not always the case, several countries present non-auction sales not registered) there is an obligation to fill it in for first-buyers (e.g. no gear or spatial information is provided).

Sales notes and geo-location data (esp. VMS data for $\geq 12\text{m}$) constitute the two major other data sources considered. In a larger part, they are used to complete or to check information coming from logbooks/coastal logbooks. Sales notes data is used to estimate landings value and to check landings weight by species and geo-location data is used to check spatial information. Such uses should be promoted and could be used even more not only to check

information but to further consolidate/improve data on for example species composition using sales notes data.

Finally, several countries integrate complementary data (e.g. *fishing licenses, geo-location data trials for specific fisheries, ...*) to complete/enhance fishing information available. This constitute another way to improve data quality.

Algorithms/methods to cross-validate/combine the different types of declarative data available to consolidate/optimize national fishing activity data.

1. Summarized information

4 MS of 12 do not have cross-check/validation systems in place. The most extended cross-checks are between logbooks and VMS data and/or sales notes. Also, MS focused on data quality checks. The data quality checks are implemented by comparison of registered coordinates with VMS data and logbooks information and difference in caught/landed amount by species in logbooks and sales notes. To get the correct weight/catches combination and value on trip level, the logbooks and landing declarations or the logbook and sales notes data/transfer information are combined by trip number/logbook number/combination of vessel-id and landing date. To combine the data R scripts are often used. In one case the coastal journals information is combined with logbooks data by merging the trip identifiers supplied by the data provider. The monthly days-at-sea are considered equivalent to the number of fishing trips.

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2. Description of practices and discussion on potentiality to enhance data quality

To join/combine the different type of declarative data available (*esp. logbooks and sales notes data or logbooks and VMS data*), two major approaches are favoured: **1)** the same identifiers are shared between the different types of declarative data sources and used to join them, therefore the link is straightforward or **2)** a specific algorithm is developed to combine data (*algorithms put in place have been sparsely detailed in the questionnaires*) first based on vessel information, dates (*i.e. fishing trip departure and return date, landings date, sales date and VMS date*) and eventually harbour name, species composition and/or landings weight (*total and/or by species*). Sharing algorithms between MS should be promoted to improve them as they have been more or less developed depending of the country. Nevertheless, the use of a common identifier shared between all the declarative data available remain the best manner to combine in an effective way the different data source.

Finally, several countries indicate that mainly only one data source is considered to calculate fishing activity data as it has all the information needed. In this case, implementing cross-validation methods could still be encouraged to enhance data quality (*allowing some complementary checks or data consolidation*) and completeness.

Algorithms/methods to estimate/assess the value of landings (*esp. for landings not sold at auctions*).

3. Summarized information

The sources of the value of landings are typically sales notes register or landings declarations. For some countries the sales notes cover the full fleet, for others, when part of the landings is not reported through sales notes, estimates based on e.g., average prices are made. Some MS are estimating the value of landings based on averages.

4. Description of practices and discussion on potentiality to enhance data quality

Several countries do not implement any estimation procedure and directly deduce the value of landings from the sales note data or the landing declarations. These countries indicated that information available in these sources is complete because there is an official requirement for first-buyer to declare.

Some other countries used a “global” average-based approach derived from an average price calculated by species notwithstanding the fact that in many cases value could be directly derived from sales note data once they have been well assigned to the declared fishing trip considered.

Finally, some countries developed a specific algorithm where value of landings are: **1)** issued from sales notes directly when they could be assigned to the fishing trip considered or **2)** if not, estimated on the basis of an average price per species. Such approach seems to be the most appropriate and should be encouraged.

Regarding the calculation of the average price per species to consider for the estimation, when it is needed, spatial (e.g. *landing harbour or maritime district*), temporal (e.g. *month or quarter*) and vessel characteristics (e.g. *vessel length ranges*) should be soundly considered, eventually in a hierarchical manner.

For some very specific species (e.g. *species never sold in auction market*), the use of expert knowledge or complementary information could be needed.

Appendix I – What is the type of declarative data considered to calculate/assess fishing activity data (summary and extract by country)?

Germany

4 types of data: logbooks ($\geq 10m$), landings declarations (*all*), trips register (*all*) and fishing fleet register (*all*)

Fishing fleet register = vessels register.

For $\leq 10m$, landing declarations are presented as monthly catch reports.

Denmark

6 types of data: sales notes (*all since 1987*), logbooks ($\geq 10m$ – $\geq 8m$ in Baltic since 1987), VMS data ($\geq 12m$ since 2012 – $\geq 15m$ in 2005-2011), AIS data ($> 15m$ since 2006), BlackBox data (mussel fisheries) and fishing fleet register (*all since 1987*)

AIS: mandatory for $> 15m$ vessels but also available on many smaller vessels. Increasing data coverage.

Blackbox: geolocalization data with sensor information mandatory for mussel fisheries and available for some EM trial fisheries.

Spain

3 types of data: sales notes (*all*), logbooks ($\geq 10m$) and VMS data ($\geq 12m$)

Sales notes are the only declaratory form of catches in vessels $< 10m$.

Estonia

All the information available: sales notes, logbooks, landings declaration, geo-localization data, fishing vessel register, etc. are centralized in the “Commercial fishing register”.

Fishermen are obliged by law to provide the requested information.

Finland

6 types of data: logbooks ($\geq 10m$), coastal logbooks of non-quota species (*monthly report issued from monthly fishing journal*) and of quota species (*fishing trip-level diary*) ($< 10m$), sales notes data (*all but mainly quota species*), vessel history and capacity data, discards data and incidental bycatch data.

Purchased catches reported by the first-sale buyers of fish. The data is mostly covered by the sales of the quota species only.

France

6 types of data: French fishing fleet register (*all*), European logbooks ($\geq 10m$), National monthly declarative fishing forms ($< 10m$), sales note data (*all*), Geolocation data ($\geq 12m$ VMS + requirement for specific fisheries and on-going trials for some other fisheries), and Scientific census survey of annual fishing activity calendars (*all*).

Sales note data. Landings statistics from auction markets. Do not cover all the French landings as non-auction sales could occur.

VMS devices (hourly basis, mandatory under EU regulation for over 12m' vessels also under national requirements for several specific fisheries e.g. Seine bay' scallop dredgers) and available for some trial fisheries.

Scientific census survey of annual fishing activity calendars. Exhaustive survey (vessels registered in the fishing fleet register) characterizing the inactivity or activity of all the vessels each month of the year and, in the latter case, the métiers practiced and the main fishing areas with the corresponding range of operation.

Ireland

4 types of data: VMS ($>12m$), logbooks ($\geq 10m$), sales notes ($<10m$) and complementary data (SSCF, $<15m$ vessels).

A sentinel fleet representing about 8% of the under 12m fleet provide effort and catch at daily resolution.

A skipper self-sampling programme started in 2021 where skippers report effort, catch, landings, discards, biological data at operational level.

Inshore VMS, high resolution spatial data are collected for some dredging fleets that provide effort and fishing distribution data.

Lithuania

5 types of data: Landings declarations and logbooks (all since 2019, $\geq 8m$ before), monthly declarative forms ($<8m \rightarrow 2018$), sales notes (all) and VMS data ($>15m$ for the last 3 years).

For calculation/assess fishing activity data the landings declarations and logbooks have been used for all vessel's segments since 2019. Until 31 December 2018 the vessel segment which length is $<8m$ and operated in the coastal area the monthly declarative form was used as source of data. The sales notes are obligatory for all fleet and even if catch is one kilo of any species.

Latvia

5 types of data: E-logbooks (open sea fishery, $\geq 10m$), coastal monthly logbooks (coastal fishery, $\leq 12m$), questionnaire "I-Fishery" ($<12m \rightarrow 2018$), sales notes and Latvian Fleet register.

... questionnaire "I-Fishery", which is compulsory for all enterprises.

The Netherlands

3 types of data: logbooks (all), sales notes (all) and VMS data ($\geq 12m$ since 2012- $\geq 15m$ - 2005).

Since 2005 all vessels longer than 15m are equipped with a Vessel Monitoring System (VMS) ... From 2012 all vessels longer than 12m are obliged to carry VMS.

Since 2018 vessels smaller than 12m are obliged to report electronic logbooks (e-lite). ... The logbook dataset follows the standard format and is considered completed for all other vessel lengths. This is the main source of landed value and what is used for all data provisions.

Poland

6 types of data: E-logbooks ($\geq 12m$), Paper logbooks (10-12m), Coastal logbooks ($<10m$), Sales notes (all), Fishing licences ($<10m$) and Geo-localisation system -VMS ($\geq 12m$).

Sweden

5 types of data: Logbooks ($\geq 10m$ + trawlers/seiners $<10m$ & 8-10m fishing in specific areas), Monthly coastal journals ($<10m$ other vessels), Landings declarations ($\geq 10m$ + trawlers/seiners $<10m$ & 8-10m fishing in specific areas), Sales notes (all), and VMS information ($\geq 12m$).



All vessels, 10 meters or more, are required to provide information in logbooks for all fishing activities in the sea. Vessels less than 10 meters fishing with trawls or seiners or land in another country than Sweden and vessels that are 8 meters or more and fish in ICES areas 22-28 and if the vessel has cod onboard that is caught in ICES areas 20-32 also have to fill in logbooks. Other vessels are obliged to fill in monthly coastal journals.

All vessels that fill in logbooks shall after landing of fish fill in a landings declaration.

Sales notes shall be filled in by a registered first-hand buyer after a sale has been closed.

Appendix II – The algorithm/methods used to cross-validate/combine the different types of declarative data available to consolidate/optimize national fishing activity data (e.g. bringing together fishermen logbooks and sales notes data) (summary and extract by country).

Germany

No algorithm. “Identifiers” are shared between the different types of data and used to join them.

For ≥ 10 m, “logbooks” & “landings declarations” are joined by haul number and species.

For all, “logbooks/landings declarations” & “trips register” are joined by trip number.

Denmark

Same “identifier” is shared between “sales notes” and “logbooks” and used to join the two types of data. Back in time, joint was based on vessel-id and landing date.

2001-recent years the “identifier” (logbook number) shared between the two types of data was defined by an algorithm (no detail available) but now it is given directly at the auctions.

The “logbook number” is not available for 1987-2000 data, algorithm used to join the two types of data is based on vessel-id and landing date (no detail available).

No detail about how VMS, AIS and BlackBox data are joined to “logbooks / sales notes” data.

Spain

Cross-validation of the different type of data (logbooks, sales notes and VMS) performed for the ≥ 10 m vessels. No detail of the algorithm implemented to join the different type of data.

Spain cross-validate different types of data available. An algorithm, called “consumption algorithm”, is used.

The data between logbook and sales note are crossed, to identify inconsistencies between landing declaration and sales notes. VMS system is used to consolidate the “vessel fishing effort” or “spatial information”, when this information is available.

Estonia

No algorithm implemented. All the needed information, related to commercial fishing, are directly centralized in the “Commercial fishing register”.

However, no cross-checking is done on a regular basis. Only occasionally misreporting is assessed by comparing the official logbook data to the data from national control authorities.

Finland

No algorithm implemented. The different types of fishing data available are merged via database key identifiers.

Different data can be merged via database key identifiers, e.g. vessel ID, fishing diary ID, form ID, sales ID and the like.

France

The different type of declarative data sources (European logbooks, national monthly declarative fishing forms, sales note data and geolocation data) are linked vessel by vessel following an algorithm based firstly on dates (fishing

trip return date, fishing sequence date, landings data, sales date, ...) and consolidated considering species composition and landings weight associated.

All these different data sources are cross-validated/combined in order to provide the best possible fishing statistic data.

SACROIS algorithms run by Ifremer allow to combine the different declarative data sources based firstly on dates and vessels. The possibility to sell the landings of a fishing trip during several sales' operation is considered also the contrary i.e. the possibility to sell during a day the landings of several fishing trips. Species composition and landings weight associated are considered to assess/strengthen the links especially between fishermen declarative and sales notes data.

Ireland

No cross-validation performed, only one data source (<10m sales notes & ≥10m logbooks) considered by vessel-length ranges.

For each vessel length category, we use only one data source: for <10m sales notes and for ≥10m logbooks.

Lithuania

Automatic crosscheck algorithm put in place to join the different type of data (landings declaration and sales notes). No detail of the algorithm implemented.

National Fisheries Data Information system (FDIS) automatically crosscheck landing declarations (before 1 January 2019 data from declarative form as well) with the sale notes species volume. The obtained discrepancy causes are investigating and looking for issue solving.

In Lithuania data base the cross-validate is established for cross checks between the sales notes and logbooks volume of species. Obtained discrepancy causes are investigating and looking for the issue solving. The main focuses of the cross-validation are on fixing the primary data.

Latvia

No formal cross-validation tool implemented. E-logbooks (≥10m) and coastal monthly logbooks (<10m) are the main data sources considered. Sales notes data used to adjust average price. VMS data used to compare with registered coordinates. Comparison catches and landings declaration in E-logbooks

E-logbooks and coastal monthly logbooks are registered in Latvian Fisheries Integrated Control and Information System (LFICIS) which is synchronised with Latvian Fleet register. In the system many of cross-checks are implemented, like: comparison of registered coordinates with VMS data, difference in caught and landed amount by species and other. Sales notes are used to adjust the average price provided by CSB if it's necessary.

The Netherlands

Logbooks and sales notes data are matched (for quality checks only) following an algorithm considering first vessel ID, dates and harbour. Species composition, total weight and weight by species is also examined to consolidate the linkage. Logbooks and VMS data are linked using the vessel identifier and date-time stamp.

Logbooks and sales notes data: the two data sources are matched by vessel ID, date and harbour and if the conditions are met a trip number from the logbooks is assigned. To ensure the right trip number is assigned to each sales note the species composition, the total weight, and the weight by species is examined. When the conditions (quality thresholds) are not met, the sale note does get assigned a trip number automatically and a manual examination of the data takes place. These quality checks are in place for internal use and the sales notes are not used for any data provision.



Logbooks and VMS: the methodology for cross checking the logbooks and VMS data is described in <https://edepot.wur.nl/248628> (Appendix B). Link VMS and logbook data • VMS and logbook datasets are linked using the vessel identifier and date-time stamp. In other words, records in the VMS dataset that fall within the departure-arrival timeframe of a trip described in the logbook are assigned the trip number from the logbook record.

Poland

Logbooks (coastal logbooks, paper logbooks and electronic logbooks) are the main data sources considered. No formal cross-validation with sales notes or complementary data (VMS data, national reference lists, vessels' patterns and fishing licences). All these data sources are used primarily to validate logbooks data.

Data from vessels under 12m are validated with national reference lists, vessels' patterns and fishing licences. Data from vessels $\geq 12m$ are validated with VMS data and national reference lists.

Sweden

All the different declarative data available (logbooks, monthly coastal journals, landings declaration, sales notes and VMS information) are combined with a unique identifier (trip id) and vessel information.

All of the documents above and VMS information is combined with a unique identifier (trip id).

These datasets all contain vessel information and trip identifier that SLU uses to combine them.

On some occasions information in landing declarations is merged in using trip identifiers supplied by SwAM in the data.

Appendix III – The algorithms/methods used to estimate/assess the value of landings (esp. for landings not sold at auctions) (summary and extract by country).

Germany

No Algorithm. Value of landings directly issued from the landing declaration.

All landings are presented in the landing declaration.

Denmark

No algorithm. Value of landings directly issued from the sales notes (landing declaration for BMS landings).

All sales are recorded in the sales notes register. For BMS information received from landing declaration.

Spain

No algorithm. Value of landings directly issued from the sales notes.

Sales notes are the only available source of value data, so no data cross-validation/combination is needed.

Estonia

No algorithm. No value provided?

Not done.

Finland

Average based approach (*average prices calculated*) used to estimate the value of quota species. Estimated by sample for non-quota species.

Notwithstanding the value of a particular landing could be calculated in some cases directly via merging SN (sales notes) and the source data (LB (logbook) and CLBQ (coastal logbook of quota species)), we utilize an average based approach. ... We use this approach because the coverage of the SN data is not (at least not yet) good enough for merging each landing with its corresponding first-sale event.

Therefore, value of landings in CLB (coastal for non-quota species) (also for some quota species) must be estimated by leveraging external data sources, i.e. purchase information obtained from a sample of 20-30 enterprises collected by the Natural Resource Institute of Finland (Luke).

France

Value of landings are estimated through a specific algorithm based on sales note data. For trip associated, value is directly deducted from sales notes and for other value is estimated through an average price or an estimated price based on expert knowledge for species with no sales available.

SACROIS algorithms include a specific algorithm to estimate the value of landings by species based on existing sales note data (sometimes directly deducted from them) or on an average price' estimation. For some fleet segment, estimated price based on expert knowledges is also used.

*... For the other landings (non-auction market sales), an average price by commercial species is assessed from sales note data by "day*landing harbour*fleet segment" considering eventual dynamic hierarchical aggregation: "day->month->quarter->year" or "landing harbour-> maritime district -> region -> seaboard". When no sales for a specie during a*

year on a seaboard raised then estimated price based on expert knowledges are considered. For abroad landings, vessel maritime district registration could be considered in replacement of landings harbour.

Ireland

Estimated value for each logbook declaration based on an average price (€/kg) calculated by species and other parameters.

The procedure for calculating these average values is hard-coded into the system and is not considered very accurate. This system of allocating values is currently being improved by the national control agency (SFPA) to better account for outliers and variability.

Lithuania

Value of landings based on the sale notes data.

There is a link between fishing trip or declarative form and specific sales note. The discrepancies of value are showing in separate report and forward for fixing issue.

Latvia

Value of landings based on an average price calculated by Central Statistical Bureau (based on the compulsory questionnaire “I-Fishery”) and eventually adjust with sales notes. Also the value provided by first purchases is used.

Landings in value – average price, calculated by Central Statistical Bureau, based on the questionnaire “I-Fishery”, which is compulsory for all enterprises.

Sales notes are used to adjust the average price provided by CSB if it's necessary.

In LFCIS system the Report of First Purchases is available where is possible to trace the sold fish up to the logbook.

The Netherlands

Value of landings based on the sale notes data.

Vessels are only allowed to sell to registered buyers at registered auctions.

Poland

Value of landings based on average price per species calculated from sales notes data. For economic data call, average price calculated consider: year and month, port of landing, species and vessel length group (>12m and >=12m).

Value of landings for economic data call is estimated based on averages, calculated taking into account:

- Year and month
- Port of landing
- Species
- Length group (<12m and >=12m)

Value of landings for RDB/RDBES and FDI data calls is estimated based on annual average price per species.

Data on fish prices comes from sales notes.

Sweden

Value of landings are 1) issued from sales notes directly when they could be assigned to the fishing trip considered or 2) estimated on the basis of a monthly average price per species calculated from sales notes data via an algorithm or directly provided by SwAM (price matrix calculation using spatial, temporal and vessel information).

Value by trip (for logbook data) are extracted from matching sales notes using trip identifiers supplied by SwAM. For trips (logbooks data) and coastal journals without matching sales notes, values are assigned based on monthly averages supplied by SwAM or aggregated directly from sales note data.

Sales directly to consumers from the fishermen is not required to report, for landings without sales notes SwAM calculates the value using a price matrix. The price matrix estimates average prices using spatial, temporal and auxiliary information regarding the vessel.

*When sales records do not exist for certain trips, the value is estimated based on an algorithm. ... Values (by usage/treatment/size class for some species) from matching trips or matching vessel-months from unique subdivisions and gear types are aggregated and used to assign values to fishing events in hierarchical order; by vessel * month, by month * region * fleet, by quarter * region * fleet, by year * region * fleet and finally by year. For some species, typically those for which mainly roe is landed or wrasses sold live, fixed mean values are supplied by SwAM.*